

Day1:

Module 1: AB-100 Certification Overview

1. What is AB-100?

The Microsoft AB-100 Accelerate Agentic AI Business Solutions certification focuses on designing, implementing, and managing enterprise-grade Agentic AI solutions using the Microsoft AI ecosystem.

This certification validates skills in:

- AI agents
- Multi-agent orchestration
- Microsoft Copilot Studio
- Azure AI Foundry
- Microsoft Graph integration
- Enterprise automation
- AI governance & security
- Autonomous workflows
- Enterprise AI architecture

2. Skills Measured in AB-100

Core Domains

Domain	Explanation
Agentic AI Fundamentals	Understanding AI agents & autonomous systems
Copilot Studio	Building enterprise copilots
AI Orchestration	Coordinating multiple agents
Azure AI Services	Using Azure OpenAI & AI Foundry
Microsoft Graph	Integrating enterprise data
AI Security & Governance	Responsible AI implementation
Multi-Agent Solutions	Designing collaborative AI systems
Enterprise Automation	AI-driven workflows

3. Exam Structure

Typical Exam Pattern

Section	Focus Area
Multiple Choice	Concepts & architecture
Scenario-Based Questions	Real-world AI design
Case Studies	Enterprise problem solving
Hands-On Labs	AI implementation
Architecture Design	AI workflow planning

4. Enterprise AI Solution Scenarios

Example Enterprise Use Cases

HR AI Assistant

- Employee onboarding
- Leave automation
- Policy answering

IT Service Desk Agent

- Incident management
- Ticket summarization
- Root cause analysis

Finance AI Agent

- Invoice processing
- Fraud detection
- Financial reporting

Healthcare AI Assistant

- Patient scheduling
- Medical document summarization
- Clinical workflow automation

5. Microsoft AI Ecosystem Overview

Core Microsoft AI Components

Service	Purpose
Microsoft Copilot Studio	Build AI copilots
Microsoft Azure OpenAI	GPT models
Microsoft Azure AI Foundry	Agent orchestration
Microsoft Power Automate	Workflow automation
Microsoft Dataverse	Enterprise data
Microsoft Graph API	Microsoft 365 integration
Microsoft Teams	AI collaboration platform

Module 2: Introduction to Agentic AI

1. What is Agentic AI?

Agentic AI refers to AI systems capable of:

- Making decisions
- Planning tasks
- Using tools
- Taking actions autonomously
- Collaborating with other agents
- Learning from context

Unlike traditional chatbots, Agentic AI systems can:

- Reason
 - Plan
 - Execute
 - Adapt
-

2. AI Agents vs Traditional Bots

Traditional Bot	AI Agent
Rule-based	Goal-based
Fixed workflows	Dynamic reasoning
Limited responses	Autonomous planning
Static decision making	Contextual intelligence
No memory	Can use memory/context

Demo 1 — Traditional Bot vs AI Agent

Objective

Understand the difference between:

- Static chatbot
- Intelligent AI agent

Demo Steps

Part A — Traditional Bot

Step 1

Open:

- [Microsoft Copilot Studio](#)

Step 2

Create:

- Simple FAQ bot

Step 3

Add predefined questions:

- "What are office timings?"
- "How to apply leave?"

Step 4

Test:

- Bot gives static responses
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Part B — Agentic AI

Step 1

Enable:

- Generative AI responses

Step 2

Connect:

- Knowledge source
- SharePoint documents

Step 3

Ask:

"I need leave policy for maternity leave and escalation process"

Step 4

Observe:

- AI reasons across documents
 - Generates contextual response
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Learning Outcome

Students understand:

- Static vs intelligent systems
 - Context awareness
 - Generative reasoning
-

3. Autonomous AI Systems

An autonomous AI system can:

- Decide next actions
- Trigger workflows
- Interact with APIs

- Coordinate tasks
 - Handle exceptions
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Real-World Example

AI Travel Assistant

User says:

"Book my Bangalore to Delhi trip next Friday"

The AI agent:

1. Checks calendar
2. Finds free schedule
3. Searches flights
4. Suggests options
5. Books ticket
6. Sends confirmation

Without manual intervention.

Demo 2 — Autonomous Workflow

Objective

Build AI workflow automation.

Demo Steps

Step 1

Open:

- [Power Automate](#)

Step 2

Create:

- Automated cloud flow

Step 3

Trigger:

- When email arrives

Step 4

Add AI Action:

- Summarize email using AI Builder/OpenAI

Step 5

Add decision:

- If urgent → create Teams alert

Step 6

Add final action:

- Generate task automatically
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Outcome

Students learn:

- AI-driven automation
 - Decision orchestration
 - Autonomous execution
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4. Task Agents vs Autonomous Agents

Task Agent

Performs:

- One specialized task

Examples:

- Email summarizer
 - Invoice extractor
 - FAQ bot
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Autonomous Agent

Performs:

- Multi-step reasoning

- Planning
- Decision-making

Examples:

- AI Operations Manager
 - AI Financial Advisor
 - AI Incident Resolver
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Demo 3 — Task Agent vs Autonomous Agent

Task Agent Demo

Build:

- PDF summarizer

Input:

- Upload policy PDF

Output:

- Summary only
-

Autonomous Agent Demo

Build:

- Enterprise HR assistant

Agent Capabilities:

- Read employee policy
 - Approve leave
 - Trigger workflow
 - Notify manager
 - Update HR system
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Learning Outcome

Students understand:

- Narrow AI tasks

- End-to-end intelligent orchestration
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5. Multi-Agent Systems

A multi-agent system contains:

- Multiple AI agents
 - Specialized responsibilities
 - Collaborative orchestration
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Example Architecture

Agent	Responsibility
HR Agent	Employee operations
Finance Agent	Payroll
IT Agent	Ticket handling
Security Agent	Compliance validation

Multi-Agent Workflow Example

Employee joins company:

1. HR agent creates profile
2. IT agent provisions laptop
3. Security agent assigns MFA
4. Finance agent activates payroll

All agents collaborate automatically.

Demo 4 — Multi-Agent Workflow

Objective

Build collaborating agents.

Demo Steps

Step 1

Open:

- [Microsoft Copilot Studio](#)

Step 2

Create:

- HR Agent
- IT Agent
- Finance Agent

Step 3

Configure Topics:

- Employee onboarding

Step 4

Create Power Automate flow:

- Trigger agents sequentially
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Step 5

HR Agent Tasks

- Collect employee data

Step 6

IT Agent Tasks

- Generate email ID
- Assign laptop

Step 7

Finance Agent Tasks

- Create payroll entry
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Outcome

Students understand:

- Agent collaboration

- Workflow chaining
 - Enterprise orchestration
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6. AI Orchestration Concepts

AI orchestration means:

- Managing AI workflows
 - Coordinating agents
 - Tool invocation
 - Memory handling
 - API integration
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Core Components of AI Orchestration

Component	Purpose
Planner	Determines next action
Memory	Stores conversation context
Tools	External APIs/functions
Executor	Executes tasks
Evaluator	Validates results

Demo 5 — AI Orchestration with Tools

Objective

Connect AI agent with APIs and workflows.

Demo Steps

Step 1

Create agent in:

- [Microsoft Copilot Studio](#)

Step 2

Add tool:

- Weather API
- Microsoft Graph API

Step 3

User asks:

"Schedule meeting tomorrow if weather is good"

Step 4

Agent Workflow

Agent:

1. Calls weather API
 2. Checks forecast
 3. Reads Outlook calendar
 4. Finds free slot
 5. Creates meeting
-

Step 5

Observe orchestration:

- Tool calling
 - Context reasoning
 - Action execution
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Day 1 Hands-On Labs

Lab Description

Lab 1 Build first Copilot agent

Lab 2 Create AI-powered FAQ assistant

Lab 3 Autonomous email processing

Lab 4 Multi-agent onboarding workflow

Lab Description

Lab 5 Microsoft Graph integration

Day 1 Mini Case Study

Enterprise Scenario

Company Problem

HR onboarding takes:

- 5 days
 - Multiple approvals
 - Manual coordination
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AI Solution

Build:

- Multi-agent onboarding system

Agents:

- HR agent
- IT provisioning agent
- Finance agent
- Security compliance agent

Module 3: Microsoft AI Ecosystem

1. Azure AI Foundry

What is Azure AI Foundry?

Microsoft Azure AI Foundry is Microsoft's enterprise AI development platform used to:

- Build AI applications
- Deploy AI models
- Manage AI agents

- Create RAG solutions
- Orchestrate enterprise AI workflows
- Manage AI governance & monitoring

It acts as the central AI development hub for:

- Generative AI
- Agentic AI
- Multi-agent orchestration
- Enterprise copilots

Key Features of Azure AI Foundry

Feature	Purpose
Model Catalog	Access GPT, Phi, Llama, Mistral models
Prompt Flow	AI workflow orchestration
AI Agents	Autonomous AI workflows
Vector Search	RAG implementation
Evaluation Tools	AI response testing
Safety Filters	Responsible AI
Monitoring	AI observability

Real-Time Example

Enterprise HR AI Assistant

Using Azure AI Foundry:

- Upload HR policies
- Create vector index
- Deploy GPT model
- Build HR AI agent
- Enable document grounding

Employees can ask:

“What is maternity leave policy?”

AI answers from enterprise documents.

Demo 1 — Create Azure AI Foundry Workspace

Objective

Create enterprise AI development environment.

Demo Steps

Step 1 — Open Azure Portal

Go to:

[Microsoft Azure Portal](#)

Step 2 — Search Service

Search:

- “Azure AI Foundry”

Click:

- Create

Step 3 — Configure Workspace

Fill:

Setting	Value
Subscription	Select subscription
Resource Group	Create new
Workspace Name	AB100-AI-Foundry
Region	East US / Sweden Central
Pricing Tier	Standard

Step 4 — Create Supporting Services

Enable:

- Azure OpenAI
 - AI Search
 - Storage Account
 - Key Vault
-

Step 5 — Launch Workspace

After deployment:

- Open AI Foundry Studio
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Step 6 — Explore Features

Show:

- Model catalog
 - Prompt flow
 - AI agents
 - Evaluations
 - Deployments
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Learning Outcome

Students learn:

- Enterprise AI environment setup
 - AI workspace architecture
 - AI development lifecycle
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2. Azure OpenAI Service

What is Azure OpenAI?

Microsoft Azure OpenAI provides enterprise-hosted OpenAI models with:

- Security
- Compliance

- Private networking
- RBAC
- Enterprise governance

Supported Models

Model	Purpose
GPT-4o	Multimodal AI
GPT-4.1	Advanced reasoning
GPT-4.1-mini	Cost-efficient AI
Embedding Models	Vector search
DALL·E	Image generation

Enterprise Benefits

- Private enterprise deployment
- Data isolation
- Compliance
- Managed AI infrastructure
- Responsible AI controls

Demo 2 — Deploy GPT Model in Azure OpenAI

Objective

Deploy enterprise GPT model.

Demo Steps

Step 1 — Open Azure AI Foundry

Navigate:

- Deployments
-

Step 2 — Click “Deploy Model”

Choose:

- GPT-4o or GPT-4.1-mini

Step 3 — Configure Deployment

Setting	Example
Deployment Name	hr-gpt-model
Tokens Per Minute	30K
Content Filter	Default
Scale Type	Standard

Step 4 — Deploy

Click:

- Deploy

Wait for deployment completion.

Step 5 — Test Model

Open:

- Chat playground

Ask:

“Summarize enterprise leave policy.”

Step 6 — Observe AI Response

Explain:

- Prompt processing
 - Context generation
 - Token usage
-

Learning Outcome

Students learn:

- GPT deployment
 - Enterprise AI hosting
 - AI model testing
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3. Copilot Studio

What is Copilot Studio?

Microsoft Copilot Studio is a low-code AI platform used to:

- Build AI agents
 - Create enterprise copilots
 - Automate workflows
 - Connect business systems
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Core Components

Component	Purpose
Topics	Conversation flows
Knowledge	Enterprise grounding
Actions	Workflow automation
Generative AI	Dynamic responses
Variables	Context storage

Real-World Use Cases

- HR assistant
- IT support bot
- AI helpdesk
- Sales assistant
- Finance automation

Demo 3 — Create First Copilot Studio AI Agent

Objective

Build enterprise AI assistant.

Demo Steps

Step 1 — Open Platform

Go to:

[Microsoft Copilot Studio](#)

Step 2 — Create Agent

Click:

- Create Agent
-

Step 3 — Configure Agent

Setting	Example
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Name	HR Enterprise Assistant
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Description	Employee support AI
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Instructions	Assist employees professionally
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Step 4 — Enable Generative AI

Turn ON:

- Generative answers
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Step 5 — Add Knowledge Source

Connect:

- SharePoint
- PDFs

- Websites

Step 6 — Test Agent

Ask:

“How many casual leaves are allowed?”

Learning Outcome

Students learn:

- AI agent creation
 - Enterprise grounding
 - Conversational AI setup
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4. Semantic Kernel

What is Semantic Kernel?

Microsoft Semantic Kernel is Microsoft’s AI orchestration SDK.

It helps developers:

- Connect LLMs with tools
 - Build AI workflows
 - Add memory
 - Perform planning
 - Orchestrate AI functions
-

Core Concepts

Concept	Purpose
Plugins	Tool integrations
Memory	Context storage
Planner	AI reasoning
Skills	AI capabilities

Concept	Purpose
Functions	Executable tasks

Example Workflow

User asks:

“Create meeting summary and send email.”

Semantic Kernel:

1. Summarizes meeting
2. Generates email
3. Calls Outlook API
4. Sends email

Architecture Flow

User → Semantic Kernel → GPT → Plugins → APIs → Response

5. Microsoft Graph API

What is Microsoft Graph API?

Microsoft Graph API provides access to:

- Outlook
- Teams
- SharePoint
- OneDrive
- Users
- Calendars

Enterprise Use Cases

Integration	Example
Outlook	Email automation

Integration	Example
Teams	Notifications
Calendar	Meeting scheduling
Users	Employee lookup
SharePoint	Document retrieval

Example AI Workflow

AI agent:

- Reads email
- Summarizes content
- Creates Teams notification
- Schedules meeting

Using:

- Graph API
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6. Power Platform Overview

Components of Power Platform

Service	Purpose
Power Apps	Low-code apps
Power Automate	Workflow automation
Power BI	Analytics
Dataverse	Enterprise database
Copilot Studio	AI agents

Enterprise AI Integration

Example:

- Copilot receives request

- Power Automate triggers workflow
- Dataverse stores data
- Power BI visualizes insights

Module 4: Enterprise AI Architecture

1. AI Solution Architecture

Enterprise AI Architecture Layers

Layer	Purpose
User Interface	Teams / Web / Mobile
AI Agent Layer	Copilot Studio
Orchestration Layer	Semantic Kernel
AI Models	Azure OpenAI
Data Layer	Dataverse / SQL
Security Layer	Entra ID / RBAC

Architecture Example

Employee → Teams AI Assistant → Azure OpenAI → Graph API → HR Database

2. AI Business Workflows

What are AI Business Workflows?

AI-powered automated business processes.

Example Workflow

Leave Approval Workflow

1. Employee requests leave
2. AI validates balance
3. Manager notified

4. Approval generated
5. HR updated automatically

Enterprise Benefits

- Faster processing
- Reduced manual work
- Better compliance
- Improved accuracy

3. AI Governance Basics

What is AI Governance?

AI governance ensures:

- Responsible AI
- Compliance
- Ethical usage
- Transparency
- Security

Governance Components

Area	Example
Access Control	RBAC
Data Protection	Encryption
Audit Logs	User tracking
Responsible AI	Bias prevention
Compliance	GDPR

Governance Best Practices

- Human approval checkpoints

- Prompt filtering
 - Content moderation
 - Logging & monitoring
 - Data masking
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4. AI Security Overview

Enterprise AI Security

Key areas:

Area	Purpose
Authentication	Entra ID
Authorization	RBAC
Encryption	Data protection
Secrets	Key Vault
Network Security	Private endpoints

Example Threats

- Prompt injection
 - Data leakage
 - Unauthorized access
 - Malicious plugins
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Security Controls

- Managed identities
 - Content filtering
 - API throttling
 - Security monitoring
-

5. Enterprise AI Lifecycle

AI Lifecycle Stages

Stage	Description
Design	Architecture planning
Build	AI development
Test	Validation
Deploy	Production rollout
Monitor	AI observability
Improve	Continuous optimization

Enterprise AI DevOps

AI lifecycle integrates:

- CI/CD
 - Model versioning
 - Monitoring
 - Governance
-

Demo 4 — Configure AI Agent Behavior

Objective

Configure enterprise AI assistant professionally.

Demo Steps

Step 1 — Open Agent

Go to:

- Copilot Studio agent
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Step 2 — Configure Greetings

Example:

“Welcome to Enterprise HR Assistant. How may I help you today?”

Step 3 — Configure Instructions

Example:

- Respond professionally
- Use HR policy documents
- Avoid hallucinations
- Escalate sensitive queries

Step 4 — Configure Prompts

Example prompt:

“Summarize leave policy in simple terms.”

Step 5 — Configure AI Behavior

Enable:

- Generative answers
- Context awareness
- Knowledge grounding

Step 6 — Test Scenarios

Ask:

- “What is maternity leave?”
- “Escalate harassment complaint”

Observe:

- AI tone
- Responsible handling
- Grounded responses

Lab 1 — Build Enterprise FAQ Chatbot

Objective

Build enterprise HR chatbot.

Lab Architecture

Component	Technology
AI Agent	Copilot Studio
GPT Model	Azure OpenAI
Knowledge Base	SharePoint/PDF
Workflow	Power Automate

Lab Tasks

Step 1

Create AI agent

Step 2

Connect HR policy PDFs

Step 3

Deploy GPT model

Step 4

Enable generative AI

Step 5

Create FAQ topics:

- Leave policy
- Payroll
- Benefits
- Work-from-home

Step 6

Test enterprise scenarios

Expected Outcome

AI chatbot answers:

- HR questions
- Policy queries
- Employee guidance

Using enterprise knowledge grounding.

Case Study — AI-Based HR Assistant Implementation

Business Problem

Enterprise HR team faces:

- High ticket volume
- Slow employee support
- Manual onboarding
- Repetitive queries

AI Solution

Build:

- AI HR assistant

Using:

- Copilot Studio
- Azure OpenAI
- Graph API
- Power Automate

AI Capabilities

Capability	Example
FAQ Handling	Leave policies
Workflow Automation	Leave approvals

Capability	Example
Email Automation	Notifications
Employee Lookup	Graph API
Knowledge Retrieval	HR documents

Enterprise Benefits

- 24/7 employee support
- Reduced HR workload
- Faster onboarding
- Improved employee experience
- Scalable enterprise automation